

VIRAL SHASTRI

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SUMMARY

GenAI-focused Python engineer with 4.5 years shipping production systems end-to-end. At AVRL, built Hyper-Y from scratch: a Python/Flask multi-stage decision pipeline on GCP processing **100+ requests/sec** across **8 enterprise TMS (Transportation Management System) integrations**. Author of **business-os** (published npm CLI for AI coding agent workflows) and the companion **business-os-genai** - a multi-provider Python agent (**LiteLLM: Ollama + Llama 3.1 / AWS Bedrock + Claude Haiku**) that generates new workflow skills via **LangChain**-orchestrated semantic retrieval on a **FAISS** store with self-repair loops. Tech-led **3eco Systems** (Series A, \$17M) across Node.js/Express, MongoDB, React, and Flutter.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Node.js
AI / GenAI: AWS Bedrock (Claude, Titan), LiteLLM, Ollama (Llama 3.1), LangChain, Hugging Face (sentence-transformers), RAG, Embeddings, FAISS, Prompt Engineering, AI Agent Orchestration
Backend: FastAPI, Flask, Express.js, REST APIs, Event-Driven (Cloud Tasks + GAE Workers)
Databases: PostgreSQL, MongoDB, Redis
Cloud & DevOps: AWS (Bedrock, Lambda, API Gateway, S3, IAM), GCP (App Engine, Cloud Storage, Cloud Run), Docker, GitHub Actions, CI/CD

EXPERIENCE

AVRL Austin, Texas, USA (Remote)
Senior SDE - Hyper-Y Jan 2024 - Present

- Designed and built Hyper-Y from scratch (reporting directly to CTO): a production decision pipeline for freight pricing processing **100+ requests/sec** across **8 enterprise TMS integrations** (BlueYonder, MercuryGate, Navisphere, UberFreight, E2Open) on GCP with Redis caching and PostgreSQL.
- Architected a multi-stage event-driven pipeline covering bid eligibility evaluation, dynamic rate calculation, and rate adjustments; configuration-driven design enabled rule changes across **10+ carrier setups** without modifying the codebase or triggering a deployment.
- Deployed on Google App Engine with Docker-based builds and GitHub Actions CI/CD; built the auth service and a structured logging pipeline capturing all bid decisions to Firestore.

AVRL Austin, Texas, USA (Remote)
RPA Automation Engineer Nov 2021 - Dec 2023

- Engineered 100+ RPA bots deployed on customer-dedicated VMs via AVRL FIT (Electron app), automating end-to-end carrier bidding across **20 TMS platforms**: bots parsed TMS interfaces, extracted shipment data, routed it through AVRL's pricing APIs, and placed bids autonomously, processing **1,000+ loads per customer per day**.
- Reduced per-bid processing time from **15 minutes (manual) to under 30 seconds**; owned the team's most complex, highest-volume TMS integrations.

ADDITIONAL EXPERIENCE

Freelance / Contract Upwork · 3eco Systems (Remote)
Lead Full-Stack Developer

- Led development of an Instagram analytics SaaS (Next.js, TypeScript, Supabase, Inngest) for a US client; system tracks accounts with **up to 1M followers** across **1,000 active users**.
- Tech lead for **3eco Systems** (Series A, \$17M, EV last-mile logistics serving Amazon, Flipkart, and BigBasket); led engineering across their Node.js/Express backend, MongoDB, web frontend, and Flutter mobile app.

AI / GENAI PROJECTS

business-os-genai github.com/darkov19/business-os-genai
Multi-Provider Decision-Routing Agent for AI Skill Generation 2026

- Python companion to my published **business-os** npm package: a decision-routing agent for AI skill generation. Multi-provider LLM gateway via **LiteLLM: Ollama + Llama 3.1** locally, **AWS Bedrock + Claude Haiku** in cloud, Groq as fallback. **sentence-transformers** + Bedrock **Titan Embeddings**, FAISS vector store, **LangChain** orchestration.
- Multi-step agent loop: route → retrieve → generate → validate → self-repair → degraded fallback. Generates structured multi-file output (SKILL.md, openai.yaml, workflow-map.md) with **Pydantic** schema validation and a max-3 retry repair loop. Deployed to **AWS Lambda + API Gateway**.

business-os npmjs.com/package/business-os
Open-source npm CLI - AI Agent Orchestration 2026

- Published Node.js CLI for orchestrating AI coding agent workflows (**Claude Code, Codex**); ships reusable agent context/instruction files, profile-based scaffolding (SaaS, D2C, partner rollout), and post-install **doctor/update** lifecycle commands. Pattern of agent-context-as-code aligns with multi-agent orchestration workflows.

EDUCATION

Master of Computer Applications (MCA) 2022
MIT World Peace University, Pune, India GPA: 9.6/10.0